Section 6.3 — Circles

Math 103: College Algebra

Basic Equation of a Circle

On a separate sheet of paper, sketch each circles described below.

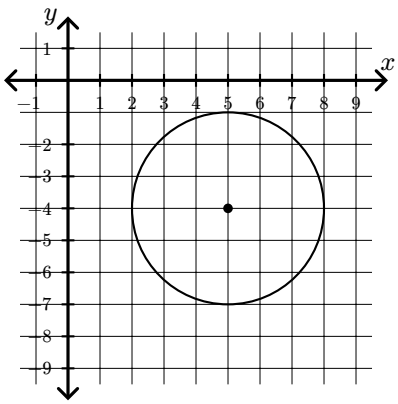
1. Centered at $(-2, 1)$ with radius 4
2. Centered at $(2, 3)$ with radius 1

Give an equation of each circles described below.

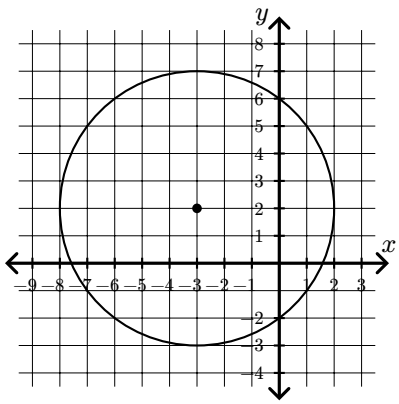
3. Centered at $(3, 5)$ with radius 2
4. Centered at $(-5, -2)$ with radius 6

Give an equation of the circle shown in each graph below.

5.



6.



Give the center and radius of the circle described by each equation below.

7. $(x-5)^2 + (y+2)^2 = 49$
8. $(x+6)^2 + (y+1)^2 = 40$

On a separate sheet of graph paper, sketch the graph of each equation below.

9. $(x+2)^2 + (y-3)^2 = 25$
10. $(x+4)^2 + (y+2)^2 = 3$

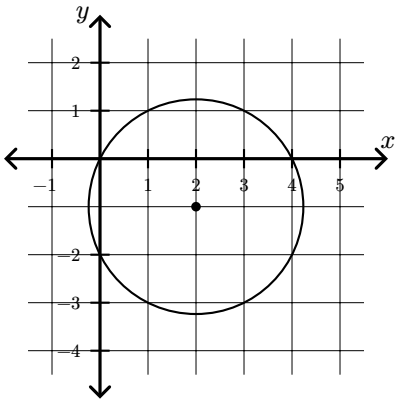
Writing Equations of Circles

Give the equation of each circle described below.

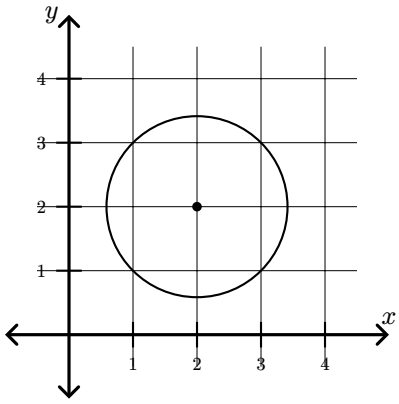
11. Centered at $(4, 1)$, through $(8, -4)$
12. Centered at $(2, -1)$, hrough $(-5, -2)$
13. Diameter has endpoints $(-10, 3)$ and $(-4, 7)$
14. Diameter has endpoints $(-6, -5)$ and $(-2, 2)$
15. Centered at $(-2, 4)$, tangent to $x = 5$
16. Centered at $(3, -5)$, tangent to $y = -2$
17. Centered at $(-5, 1)$, tangent to the x-axis
18. Centered at $(-4, -3)$, tangent to $y = -\frac{5}{3}x - 21$

Give an equation of the circles graphed below.

19.



20.



Working with Standard Form

Give an equation of each circle below in standard form.

21. Centered at $(4, -2)$ with radius 3
22. Centered at $(-2, -5)$ through point $(-1, -2)$
23. Diameter has endpoints $(-2, -1)$ and $(5, 3)$
24. Centered at $(2, 1)$ and tangent to $y = 4$

Rewrite each equation below in center-radius form. If the equation describes a true circle, give its center and radius. If it doesn't, state whether it is the equation of a degenerate circle (single point), or if the equation has no solution.

25. $x^2 + y^2 - 4x - 6y + 9 = 0$
26. $x^2 + y^2 - 2x + 6y - 15 = 0$
27. $4x^2 + 4y^2 - 8x + 16y + 11 = 0$
28. $x^2 + y^2 + 10x + 9 = 0$
29. $x^2 + y^2 - 6x - 10y + 34 = 0$
30. $x^2 + y^2 + 4x - 8y + 11 = 0$
31. $x^2 + y^2 + 10x - 10y + 54 = 0$
32. $x^2 + y^2 - 2x + 10y + 26 = 0$

Rewrite each equation below in center-radius form. Then, graph the circle on a separate sheet of graph paper.

33. $x^2 + y^2 - 4y - 12 = 0$
34. $x^2 + y^2 + 10x + 2y + 17 = 0$
35. $4x^2 + 4y^2 - 8x + 28y + 17 = 0$